From Low Lunar Orbit to Robotic Berthing

An End-to-End Numerical Simulation of a Crewed Return Phase

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Build of 5 September 2026

Abstract

An ascent stage leaves the lunar surface, reaches a 100 km circular orbit with the dispersion that a single-string engine and a migrating centre of mass inevitably leave behind, and must reach a mothership 300 km higher and 10° ahead. This report documents an end-to-end numerical simulation of that return phase, written from first principles in MATLAB using only `ode45` and `fminsearch`.

Six coupled studies are presented. A Hohmann transfer between the two circular equatorial orbits is designed and phased, costing 118.80 m/s after a 9.44 h wait for the launch window; the propagated trajectory is validated against a closed-form Kepler solution to 0.191 mm over the full 10.55 h timeline. Relative motion at the rendezvous point is treated with the Hill–Clohessy–Wiltshire state transition matrix: an injection error of 529.0 m and 0.8796 m/s grows into a 20.99 km drift within three orbits, is removed for 1.123 m/s onto a 50 m V-bar hold, and the vehicle is then walked to the docking port for a further 0.1567 m/s. The same manoeuvre plan flown against a Cowell model with lunar J_2 and the Earth as a third body, deliberately without retargeting, misses the rendezvous by 15.14 km, of which the Earth contributes only 115.7 m; the corresponding mid-course correction budget is of order 10.7 m/s. Re-converging the transfer by single shooting in the Earth–Moon circular restricted three-body problem closes the rendezvous to 4.8×10^{-8} m for a correction of 8.84 mm/s, confirming that a low lunar orbit sits deep enough in the lunar well for Earth tides to be operationally negligible. Finally a five-revolute free-flying manipulator of 7.7 m reach is characterised: forward kinematics, the twist-propagation matrix, kineto-static duality giving joint torques up to 396.5 N m under a 100 N berthing load, and a weighted damped least-squares inverse kinematics converging to 0.99 mm in 18 iterations.

Every headline figure is cross-checked, either against a closed-form re-derivation from the raw constants or against an independent implementation of the same mission geometry. The verification results, including the two quantities that differ and why, are reported rather than smoothed over.

Keywords: lunar rendezvous, Hohmann transfer, orbital phasing, Clohessy–Wiltshire, Cowell propagation, J_2, circular restricted three-body problem, space manipulator, kineto-statics, damped least squares.

Contents

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1 Introduction

1.1 The problem

Every crewed lunar architecture that has flown, and every one currently proposed, splits the vehicle in two. A descent stage carries the landing gear, the throttleable engine and the surface consumables, and it stays where it lands. Only an ascent stage returns to orbit. The reason is arithmetic: mass that does not have to be re-accelerated to lunar orbital speed does not have to be carried from Earth in the first place. Apollo used this split six times, and the Artemis Human Landing System requirements inherit it [5].

The consequence is that the return leg is a rendezvous problem, and a harder one than the launch. The ascent stage flies on a single engine with no aerodynamic stabilisation, no throttle authority, a centre of mass that migrates as the tanks drain, and an inertia tensor with significant off-diagonal terms. Its guidance can steer the thrust vector and choose when to stop, but it cannot trim the acceleration profile. Whatever mismatch remains between predicted and actual performance appears directly as a velocity error at cut-off. From orbit, an ascent stage is completely described by the dispersed state it delivers, and everything after that point is somebody else's propellant budget.

This report follows that budget from the parking orbit to the moment a robotic arm closes on a grapple fixture.

1.2 Scope and mission definition

Table 1 defines the case study. Both vehicles are on circular, equatorial, coplanar orbits about the Moon; the chaser is a lunar module (LM) at 100 km altitude, the target a mothership (MS) at 400 km, leading by $\phi_0 = 10°$ at the epoch.

Table 1: Mission definition and physical constants. Values are held in a single struct, `config/mission_constants.m`; no physical number appears as a literal anywhere else in the code base.

Quantity	Symbol	Value	Unit
Lunar gravitational parameter	μ_M	4902.8	km^3/s^2
Lunar mean radius	R_M	1737.4	km
Lunar oblateness	J_2	2.033×10^{-4}	–
Earth gravitational parameter	μ_E	3.986×10^5	km^3/s^2
Mean Earth–Moon distance	d_{EM}	384 400	km
Chaser orbit radius	R_1	1837.4	km
Target orbit radius	R_2	2137.4	km
Initial target lead	ϕ_0	10	deg
CR3BP mass parameter	μ	0.01215	–
Random seed	–	42	–

The reference frame for the orbital work is Moon-centred inertial (MCI), equatorial, with $+z$ along the lunar north pole and motion counter-clockwise in the xy-plane. At $t = 0$ the chaser is on the $+x$ axis. Proximity operations use a local-vertical/local-horizontal frame attached to the target, defined carefully in Section 3 because its handedness is the single most error-prone convention in the whole study.

1.3 What is modelled, and what is not

This is an engineering-faithful simulation of a restricted model. It is not a digital twin of any flown or planned mission, and it is worth being explicit about the boundary before presenting results.

Modelled: two-body dynamics with tangential impulsive manoeuvres; orbital phasing; linear and exact nonlinear relative motion; Cowell propagation with lunar J_2 and Earth third-body perturbation; the Earth–Moon circular restricted three-body problem; the kinematics and kineto-statics of a five-revolute manipulator on a controlled base.

Not modelled: the lunar gravity field beyond J_2, in particular the mass concentrations mapped by GRAIL [9] and the C_{22} ellipticity term; solar radiation pressure and eclipse cycling; finite-burn losses, propellant slosh and structural flexibility; orbital inclination and nodal regression; a near-rectilinear halo orbit in place of the circular target orbit; and the free-floating multi-body dynamics of the arm, whose inertia matrix and Coriolis terms are computed but never integrated. Section 8.3 returns to each of these and states what it would change.

1.4 Numerical approach and verification philosophy

Everything is implemented from first principles in MATLAB. The only solvers used are `ode45` for propagation and `fminsearch` for the three-body shooting problem; no MathWorks toolbox beyond base MATLAB is required, and no external ephemeris is read.

The organising principle of the work is that a simulation is worth exactly as much as its verification. Each part therefore closes with a verification subsection stating what was compared against what. Three kinds of check are used throughout:

(i) **Closed form.** Where an analytical solution exists it is implemented independently and differenced against the numerical one. The Kepler propagation check of Section 2.4 is the strongest example.

(ii) **Invariants.** Quantities the dynamics must conserve are monitored: specific energy and angular momentum on each Keplerian leg, and the Jacobi constant along the three-body arc.

(iii) **Independent implementation.** Selected results are compared against a separate implementation of the same mission geometry. Where the two disagree, Section 8.2 says by how much and why, rather than tuning one towards the other.

A dedicated script, `tests/audit_reference.m`, performs all of these automatically at the end of every build and writes a pass/warn/fail table. The results quoted throughout this report are the ones that script validated.

2 Orbital transfer and phasing

2.1 Hohmann geometry

The minimum-energy two-impulse transfer between coplanar circular orbits is an ellipse tangent to both [3]. With R_1 and R_2 the two radii,

$$a = \frac{R_1 + R_2}{2}, \qquad e = \frac{R_2 - R_1}{R_2 + R_1}, \tag{1}$$

giving $a = 1987.4\,\text{km}$ and $e = 0.0755$. Circular and elliptical speeds follow from the vis-viva equation,

$$v_c(r) = \sqrt{\frac{\mu_M}{r}}, \qquad v(r) = \sqrt{\mu_M\left(\frac{2}{r} - \frac{1}{a}\right)}, \tag{2}$$

and because the transfer is tangent at both ends the two impulses are purely along-track:

$$\Delta v_1 = v(R_1) - v_c(R_1), \qquad \Delta v_2 = v_c(R_2) - v(R_2). \tag{3}$$

The time of flight is half the ellipse period, $\Delta t = \pi\sqrt{a^3/\mu_M} = 3975.17\,\text{s}$, i.e. 66.25 minutes. Table 2 collects the design.

Table 2: Hohmann transfer design, $100 \to 400\,\text{km}$ circular lunar orbit. The two impulses together cost 118.80 m/s, which is the reference against which every later correction is measured.

Quantity	Value	Unit
Semi-major axis a	1987.4	km
Eccentricity e	0.0755	–
Circular speed at R_1	1.6335	km/s
Circular speed at R_2	1.5145	km/s
Ellipse speed at periapsis	1.6940	km/s
Ellipse speed at apoapsis	1.4563	km/s
Departure impulse Δv_1	60.52	m/s
Arrival impulse Δv_2	58.28	m/s
Total Δv	**118.80**	**m/s**
Time of flight	3975.17	s

2.2 Phasing: why the vehicle waits nine hours

Designing the ellipse is the easy half. The chaser may only ignite when the geometry places the target at the arrival point at the arrival time.

During the transfer the chaser sweeps exactly π radians while the target advances $n_2\Delta t$, with $n_i = \sqrt{\mu_M/R_i^3}$. Rendezvous therefore requires the target to lead the chaser at ignition by

$$\Delta\theta_H = \pi - n_2\Delta t = 18.61^\circ. \tag{4}$$

At the epoch the target leads by only $\phi_0 = 10^\circ$. Because the inner orbit is the faster one, the target's apparent lead *decreases* at the relative rate $|n_2 - n_1|$: the required 18.61° will not grow back from 10°, it has just been missed. The chaser must wait for the lead to run down through zero and all the way round:

$$\Delta\theta_{\text{req}} = 2\pi - (\Delta\theta_H - \phi_0), \qquad t_{\text{wait}} = \frac{\Delta\theta_{\text{req}}}{|n_2 - n_1|} = 33988\,\text{s} = 9.44\,\text{h}. \tag{5}$$

The implementation carries the general conditional, so a geometry in which the window has *not* been missed takes the short branch instead. Total mission duration is 10.55 h, of which the transfer itself is barely an hour. That ratio is the operationally interesting result of this section: in a rendezvous of this kind, propellant is cheap and patience is the constraint.

The rendezvous closes exactly by construction. At arrival the chaser is at $n_1 t_{\text{wait}} + \pi$ and the target at $\phi_0 + n_2(t_{\text{wait}} + \Delta t)$; substituting (5) makes the difference exactly 2π.

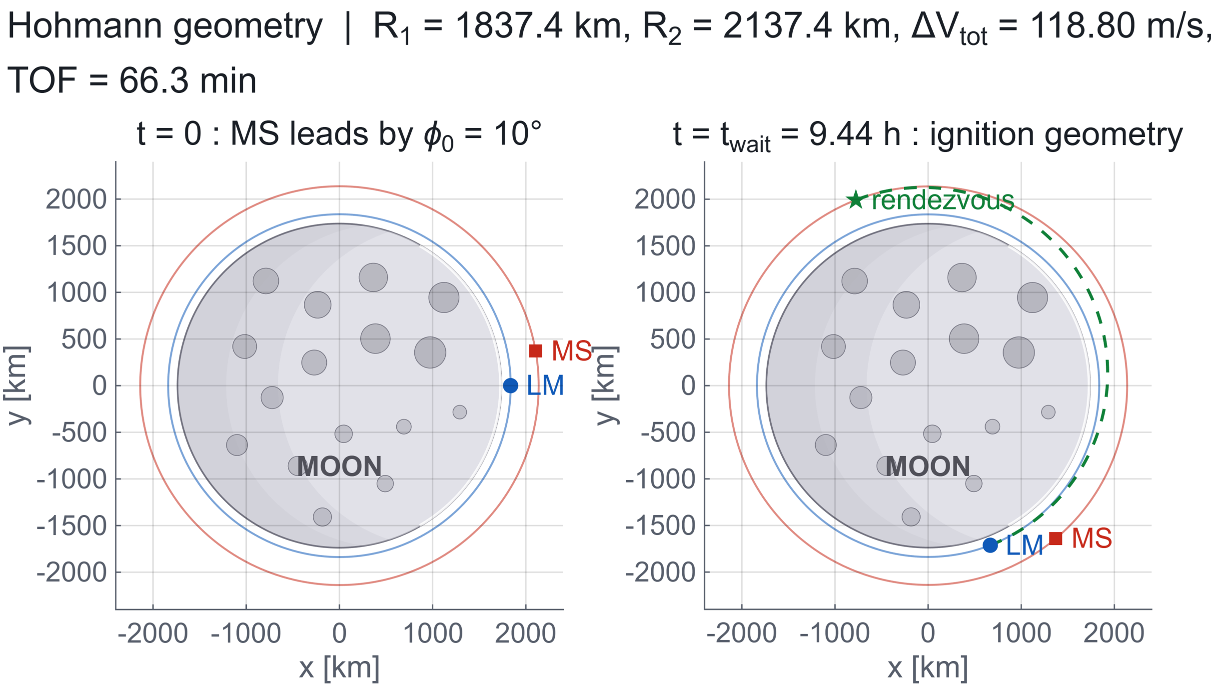

Figure 1: Transfer geometry at the epoch (left) and at ignition (right). The target leads by 10° at $t = 0$ but the manoeuvre needs 18.61°, so the chaser coasts 9.44 hours on the inner orbit before the burn. The dashed ellipse is the 118.80 m/s transfer.

2.3 Numerical propagation

The full timeline is integrated in Cartesian coordinates with the unperturbed two-body field,

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\mu_M \frac{\mathbf{r}}{\|\mathbf{r}\|^3}, \tag{6}$$

using `ode45` at relative and absolute tolerances of 10^{-12}. At $t = t_{\text{wait}}$ the departure impulse is applied along the instantaneous velocity unit vector; at apoapsis the second impulse circularises the orbit so that the constants of motion display a third plateau.

2.4 Verification against the closed-form solution

The same timeline is then evaluated analytically and differenced. Circular legs are trivial. On the transfer ellipse the mean anomaly $M = n_H(t - t_{\text{wait}})$ is inverted for the eccentric anomaly by Newton–Raphson,

$$E_{k+1} = E_k - \frac{E_k - e \sin E_k - M}{1 - e \cos E_k}, \tag{7}$$

iterated to 10^{-14}, and the perifocal state

$$\mathbf{r}_{pf} = \begin{bmatrix} a(\cos E - e) \\ a\sqrt{1-e^2}\,\sin E \\ 0 \end{bmatrix} \tag{8}$$

is rotated into MCI by $R_z(\theta_p)$, with θ_p the ignition longitude.

Over the whole 10.55 h timeline the two solutions differ by at most 0.191 mm in position, and the numerically propagated rendezvous closes to 5.6×10^{-5} m. This is the single most important number in the report: it is the licence to believe every perturbed result that follows. When Section 4 reports a 15.14 km miss under J_2, we know that 15 km is physics and not integrator drift, because the same integrator reproduces the analytical solution to a fifth of a millimetre.

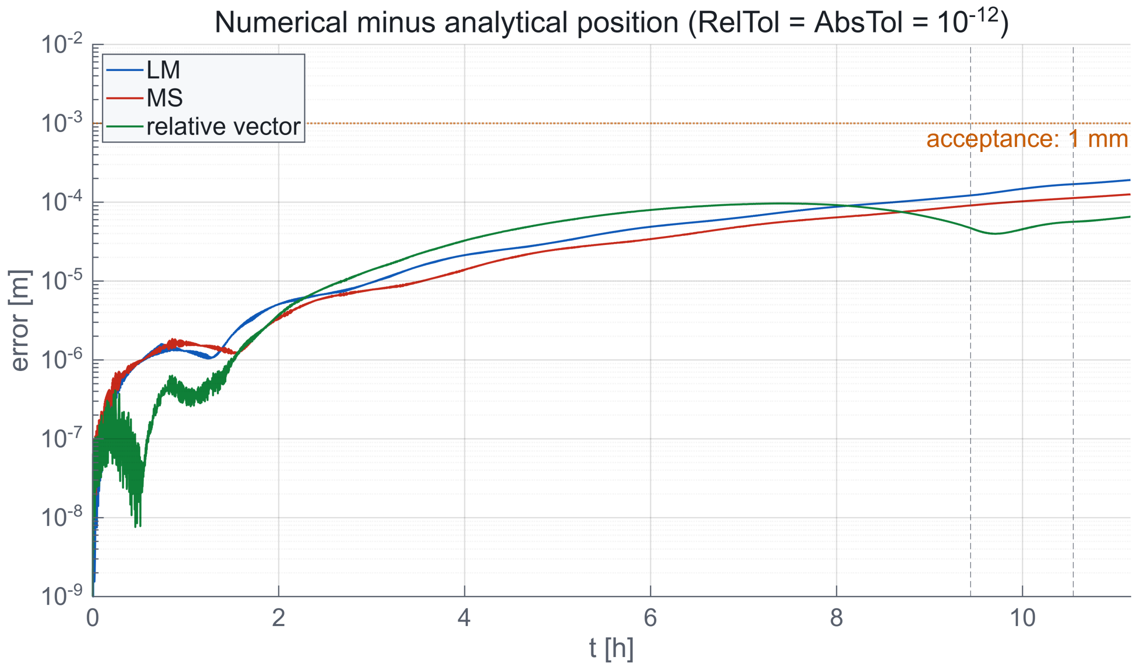

Figure 2: Difference between the `ode45` solution and the closed-form Kepler solution over the full timeline, log scale. The error grows smoothly to 0.191 mm, three orders of magnitude below the 1 mm acceptance line. Dashed verticals mark the two impulses.

2.5 Constants of motion

For a further independent check the specific energy, angular momentum and eccentricity vector are evaluated along the trajectory,

$$\varepsilon = \frac{v^2}{2} - \frac{\mu_M}{r}, \qquad \mathbf{h} = \mathbf{r} \times \mathbf{v}, \qquad \mathbf{e} = \frac{1}{\mu_M}\left[\left(v^2 - \frac{\mu_M}{r}\right)\mathbf{r} - (\mathbf{r}\cdot\mathbf{v})\mathbf{v}\right]. \tag{9}$$

These must be piecewise constant, stepping only at the two impulses. Table 3 compares the simulated plateaus with the closed-form values $\varepsilon = -\mu_M/2a$ and $\|\mathbf{h}\| = \sqrt{\mu_M a(1-e^2)}$; they agree to twelve significant figures.

Table 3: Constants of motion per leg. Analytical and simulated values agree to better than 10^{-12} relative, confirming that the impulses are applied cleanly and that no energy leaks through the integrator.

Leg	e	$\|\mathbf{h}\|$ [km^2/s]	ε [km^2/s^2]
Inner circular	0	3001.40	−1.33417
Transfer ellipse	0.0755	3112.61	−1.23347
Outer circular	0	3237.17	−1.14691

3 Relative dynamics and proximity operations

3.1 The injection error is the whole point

The rendezvous of Section 2 closes to a fraction of a millimetre, which no ascent stage has ever achieved. The dispersion listed in Section 1.1 – single-string propulsion, thrust-to-centre-of-mass misalignment, active-only attitude control, migrating inertia – arrives at the top of the climb as a position and velocity offset. This part models that offset explicitly and prices its removal.

The error is drawn once from a seeded generator: a random direction with magnitude uniform in $[500, 1000]$ m for position and $[0.1, 1]$ m/s for velocity. The realisation used throughout is $\|\delta\mathbf{r}_0\| = 529.0$ m and $\|\delta\mathbf{v}_0\| = 0.8796$ m/s. The seed, not the numbers, is the reproducibility mechanism.

3.2 Frame definition, and the sign trap inside it

Relative motion is expressed in a local frame attached to the target. This report uses

$$\hat{\mathbf{x}} = -\hat{\mathbf{v}}, \qquad \hat{\mathbf{y}} = -\hat{\mathbf{h}}, \qquad \hat{\mathbf{z}} = +\hat{\mathbf{r}}, \tag{10}$$

that is, x along the V-bar measured positive *astern* of the target, y cross-track along the negative orbit normal, and z along the R-bar, positive outward. The triad is right-handed, since $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = (-\hat{\mathbf{v}}) \times (-\hat{\mathbf{h}}) = \hat{\mathbf{v}} \times \hat{\mathbf{h}} = \hat{\mathbf{r}}$.

Two remarks justify what looks at first like a perverse choice. Operationally, V-bar range is quoted astern: a "50 m V-bar hold" is 50 m behind the target, and the approach then walks x from $+50$ m down to zero at the port. Mathematically, the price is that the target's angular velocity is $-n$ about the frame's own y axis, which flips the sign of every Coriolis term relative to the textbook radial/along-track ordering. Linearising about a circular chief [2, 4] gives

$$\ddot{x} - 2n\dot{z} = 0, \tag{11}$$

$$\ddot{y} + n^2 y = 0, \tag{12}$$

$$\ddot{z} + 2n\dot{x} - 3n^2 z = 0. \tag{13}$$

This is exactly where a sign error hides most comfortably: copy a state transition matrix written for x radial, label the axes V-bar and R-bar, and the free-drift plot still looks plausible while every targeting solution is quietly wrong. The convention was therefore settled empirically rather than asserted. Section 8.2 reports the test: fed an independent implementation's injection error, the targeting of Section 3.5 reproduces that implementation's Δv to 1.4%, whereas the mirrored convention does not.

A physical check completes the argument. Release the chaser 100 m radially outward at zero relative velocity: it is on a higher, slower orbit and must fall behind. The state transition matrix returns $x = +11.3$ km after three orbits, and since $+x$ is astern, it has indeed fallen behind. Reading $+x$ as "ahead" is the trap; the sign is correct once the axis direction is stated.

3.3 State transition matrix

With $\tau = nt$ and the state $\delta\mathbf{x} = [x\ y\ z\ \dot{x}\ \dot{y}\ \dot{z}]^\top$, equations (11)–(13) integrate in closed form to $\delta\mathbf{x}(t) = \Phi(t)\,\delta\mathbf{x}(0)$ with

$$\Phi(t) = \begin{bmatrix} 1 & 0 & 6(\tau - \sin\tau) & \frac{4\sin\tau - 3\tau}{n} & 0 & \frac{2(1-\cos\tau)}{n} \\ 0 & \cos\tau & 0 & 0 & \frac{\sin\tau}{n} & 0 \\ 0 & 0 & 4 - 3\cos\tau & \frac{2(\cos\tau - 1)}{n} & 0 & \frac{\sin\tau}{n} \\ 0 & 0 & 6n(1-\cos\tau) & 4\cos\tau - 3 & 0 & 2\sin\tau \\ 0 & -n\sin\tau & 0 & 0 & \cos\tau & 0 \\ 0 & 0 & 3n\sin\tau & -2\sin\tau & 0 & \cos\tau \end{bmatrix}. \tag{14}$$

Three properties are checked automatically: $\Phi(0) = I$ to machine precision; $\dot{\Phi}(0)$ equals the system matrix of the first-order form, which is what actually pins the Coriolis signs; and $\Phi(t)\Phi(-t) = I$.

3.4 Free drift

Propagating the injection error with no correction is instructive. Figure 3 shows the result over three target orbits: the out-of-plane component is a bounded harmonic at the orbital frequency, while the in-plane motion develops a secular along-track drift driven by the period mismatch. The separation reaches 20.99 km.

That number is the argument for active proximity operations in one line. A 0.88 m/s velocity error, roughly 0.05% of orbital speed, becomes a 21 km miss within seven hours.

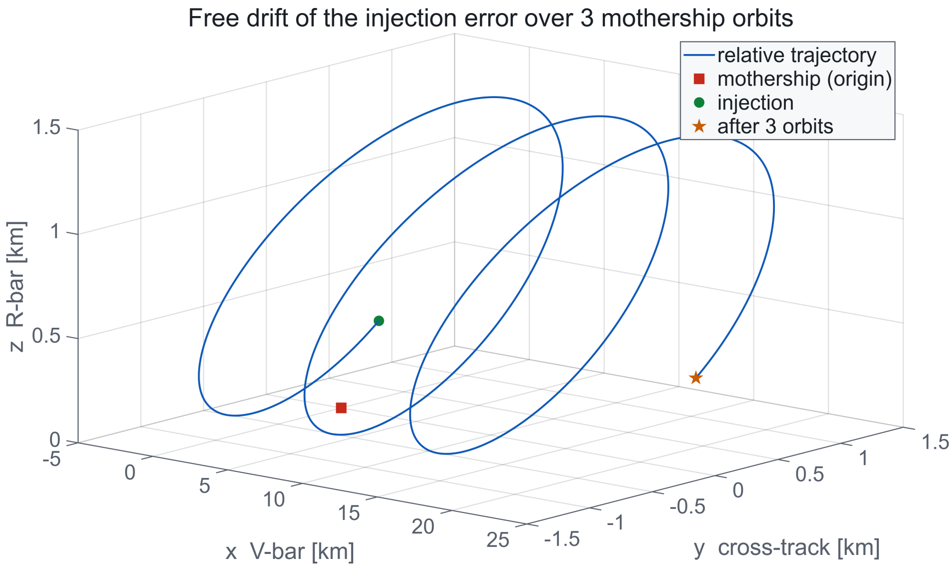

Figure 3: Uncorrected injection error over three target orbits in the local frame. Starting from 529.0 m and 0.8796 m/s, the chaser drifts 20.99 km along-track. Cross-track motion stays bounded.

3.5 Two-impulse hold acquisition

Partitioning (14) into 3×3 blocks, the transfer from the injection state to a prescribed hold point in a prescribed time is a single linear solve:

$$\mathbf{v}_0^+ = \Phi_{rv}^{-1}(\mathbf{r}_{\text{hold}} - \Phi_{rr}\,\delta\mathbf{r}_0), \qquad \Delta v_1 = \mathbf{v}_0^+ - \delta\mathbf{v}_0, \tag{15}$$

$$\Delta v_2 = \mathbf{v}_{\text{hold}} - \left(\Phi_{vr}\,\delta\mathbf{r}_0 + \Phi_{vv}\,\mathbf{v}_0^+\right). \tag{16}$$

The hold point is $\mathbf{r}_{\text{hold}} = [50, 0, 0]^\top$ m and the transfer time $0.3\,T_{MS} = 2660$ s. The cost is $\Delta v_{\text{hold}} = 1.123$ m/s, and propagating the resulting two-burn trajectory back through (14) arrives within 10^{-13} m of the target point.

The block Φ_{rv} becomes singular whenever the transfer time approaches a multiple of the orbital period, because after a full revolution the initial velocity no longer controls the final position. Its condition number is therefore checked before every solve and the transfer time nudged by 1% until the inversion is safe; at $0.3\,T_{MS}$ the reciprocal condition number is 0.21, comfortably clear.

3.6 Forced straight-line V-bar approach

From the hold point the vehicle is walked to the port in $N = 5$ equal legs of 200 s, each targeting the waypoint $\mathbf{r}_k = \mathbf{r}_{\text{hold}}(1 - k/N)$, followed by a braking impulse that nulls the residual velocity at contact. The total is $\Delta v_{\text{dock}} = 0.1567$ m/s, and the trajectory terminates 3×10^{-16} m from the origin.

This profile is not chosen for elegance. A sequence of short targeted hops keeps plume impingement away from the target, bounds the closing rate between checkpoints, and gives the crew or the ground a natural abort opportunity at every waypoint – the pattern used on real V-bar approaches [4].

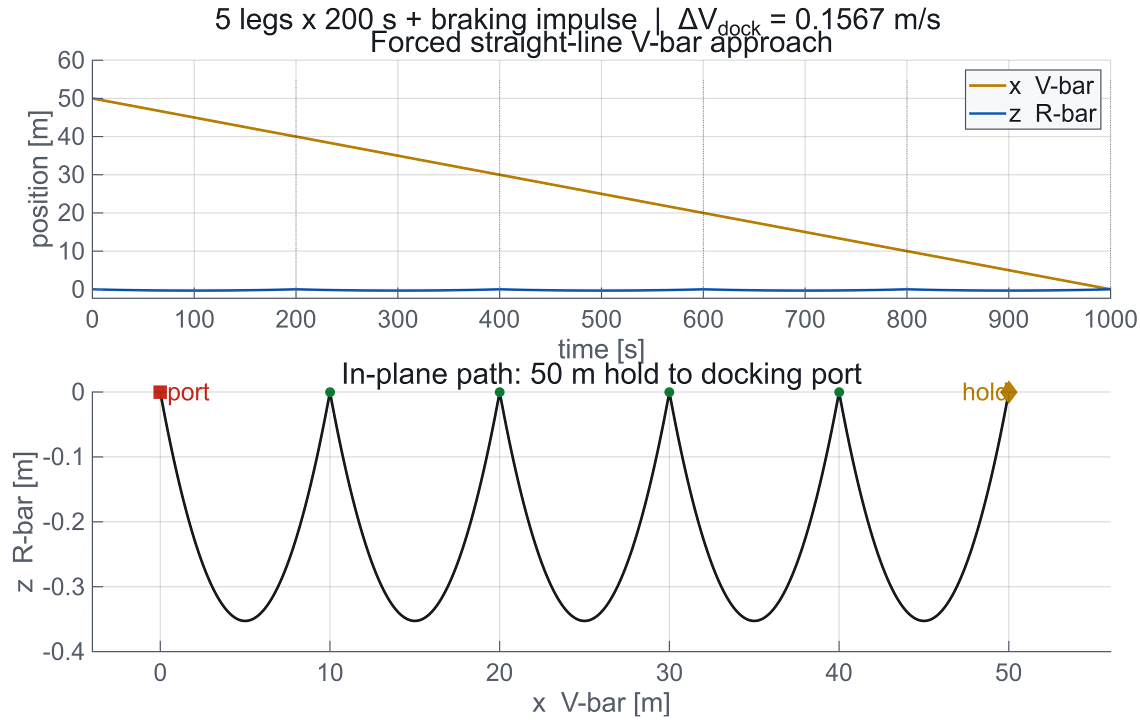

Figure 4: Forced straight-line approach. Top: V-bar and R-bar components against time, with the six impulses marked. Bottom: the in-plane path from the 50 m hold to the docking port, total cost 0.1567 m/s.

The full proximity phase therefore costs 1.280 m/s, about 1% of the 118.80 m/s transfer budget. That is the price of the ascent dispersion, and it is only obtainable by modelling both phases in the same units.

3.7 Verification against the nonlinear equations

The linear model is finally checked against the exact nonlinear relative equations of motion for a circular chief, integrated with the same tolerances. Writing $\rho^2 = x^2 + y^2 + z^2$ and $r_c = \sqrt{x^2 + y^2 + (R_2 + z)^2}$, and using the regularised form

$$q = \frac{\rho^2 + 2R_2 z}{R_2^2}, \qquad F = \frac{q\,(2 + q + \sqrt{1+q})}{(1+q)^{3/2}\,(1 + \sqrt{1+q})}, \tag{17}$$

avoids differencing two nearly equal inverse cubes, which matters when $\rho/R_2 \sim 10^{-5}$.

The two models part company by 102.7 m after three orbits at kilometre range – about half a percent of the separation. At the 50 m hold point, where $\rho/R_2 = 2.3 \times 10^{-5}$, they agree to well under a millimetre. The linearisation is used exactly where it is valid, and the divergence at long range is quantified rather than assumed away.

4 Perturbed dynamics

4.1 Purpose of the experiment

Sections 2 and 3 live in a pure two-body field. This part asks a specific operational question: if the manoeuvre plan is designed in that field and flown against a more complete one *without retargeting*, how far off does it arrive? The answer sizes the mid-course correction that a real flight plan must carry.

4.2 Cowell formulation

Perturbations are added directly to the acceleration in Cartesian coordinates,

$$\ddot{\mathbf{r}} = -\mu_M \frac{\mathbf{r}}{r^3} + \mathbf{a}_{J_2} + \mathbf{a}_{3B}. \tag{18}$$

The oblateness term is the standard second zonal harmonic [8],

$$\mathbf{a}_{J_2} = \frac{3J_2\mu_M R_M^2}{2r^4} \begin{bmatrix} \frac{x}{r}\left(5\frac{z^2}{r^2} - 1\right) \\ \frac{y}{r}\left(5\frac{z^2}{r^2} - 1\right) \\ \frac{z}{r}\left(5\frac{z^2}{r^2} - 3\right) \end{bmatrix}. \tag{19}$$

The full three-component expression is retained even though the orbits are equatorial. At $z = 0$ it collapses correctly to a purely radial correction, so special-casing it would add a branch and remove a check.

The Earth is included as a third body in Battin's difference form [1],

$$\mathbf{a}_{3B} = \mu_E \left(\frac{\mathbf{r}_E - \mathbf{r}}{\|\mathbf{r}_E - \mathbf{r}\|^3} - \frac{\mathbf{r}_E}{\|\mathbf{r}_E\|^3} \right), \tag{20}$$

with the Earth on a circular path about the Moon at the synodic rate $n_{EM} = \sqrt{(\mu_M + \mu_E)/d_{EM}^3}$. The second term is the indirect part: it removes the acceleration of the origin, because the frame is Moon-centred and the Moon is itself falling towards the Earth. Dropping it is a classic error that inflates the computed effect by orders of magnitude; retaining it is what makes the 116 m result of Section 4.5 meaningful.

4.3 Acceleration budget

Before integrating anything it is worth knowing what matters. Table 4 evaluates the characteristic magnitudes at the chaser's orbit radius.

Table 4: Characteristic accelerations at $r = R_1 = 1837.4$ km. Oblateness dominates the perturbation budget by more than an order of magnitude over the Earth, which in turn dominates radiation pressure by two.

Source	Magnitude [m/s^2]	Ratio to central
Central lunar attraction	1.452	1
Lunar J_2	3.96×10^{-4}	2.7×10^{-4}
Earth third body	2.58×10^{-5}	1.8×10^{-5}
Solar radiation pressure (estimated)	$\sim 10^{-7}$	$\sim 7 \times 10^{-8}$

Radiation pressure is estimated rather than integrated. For an area-to-mass ratio of order $0.01\,\mathrm{m^2/kg}$ and a reflectivity near 1.3, the cannonball acceleration at 1 AU is around $10^{-7}\,\mathrm{m/s^2}$. It is also intermittent: a 100 km lunar orbit has a period near two hours and spends roughly 30–40% of each revolution in shadow, so the perturbation switches on and off rather than accumulating smoothly. Integrated over the 10.55 h mission the velocity change is of order $10^{-3}\,\mathrm{m/s}$, three orders below the correction J_2 already demands. The code carries a disabled flag for it and the omission is deliberate.

4.4 Why an equatorial orbit still feels J_2

For $z \equiv 0$ equation (19) reduces to $\mathbf{a}_{J_2} = -\frac{3}{2} J_2 \mu_M R_M^2 r^{-5} \, [x, y, 0]^\top$: a purely radial correction. It does not tilt the plane and it does not regress the node. What it does is stiffen the central field by a *radius-dependent* amount,

$$\frac{\delta a_r}{a_r} = \frac{3}{2} J_2 \left(\frac{R_M}{r}\right)^2, \tag{21}$$

which is 2.73×10^{-4} at the chaser radius and 2.01×10^{-4} at the target radius. A circular orbit responds with $\delta n/n = \frac{1}{2}\delta\mu_{\text{eff}}/\mu$, so the two vehicles accumulate *different* along-track phase. Over ten and a half hours that difference is kilometres. The miss is therefore almost purely along-track, which is exactly what the simulation produces.

4.5 Results

Both vehicles are propagated over the identical Part 1 timeline with the identical perturbed dynamics; only their initial states differ. The departure impulse keeps its Keplerian magnitude but is applied along the *perturbed* velocity unit vector sampled at t_{wait}, which is what an open-loop vehicle would actually do. No retargeting is performed.

Table 5: Rendezvous miss at the nominal arrival epoch, flying the Keplerian plan against progressively more complete dynamics. The Keplerian control case through the identical code path closes to 5.6×10^{-3} m, so the residuals below are physics and not integrator noise.

Dynamics	Miss	Unit
Two-body (control case)	0.006	m
Lunar J_2	15.14	km
Lunar J_2 + Earth third body	15.02	km
Third-body contribution alone	115.7	m
Equivalent mid-course correction $\approx n_2\|\mathbf{d}\|$	10.7	m/s

Three readings follow. First, oblateness alone moves the meeting point by 15.14 km – far beyond any docking corridor, and the reason a real plan carries a correction. Second, the Earth contributes only 115.7 m of that: at 100–400 km altitude the vehicle is deep enough in the lunar well that Earth tides are a hundred times smaller than lunar oblateness. Third, removing the miss over roughly one orbit costs about 10.7 m/s, some 9% of the nominal transfer budget. That is a realistic mid-course allocation for a design of this fidelity.

4.6 Element histories

Classical elements are extracted along the perturbed trajectories with a degeneracy-safe conversion: the orbits are simultaneously circular and equatorial, so both the node vector and the eccentricity vector vanish. The implementation falls back on the true longitude, taking the node along

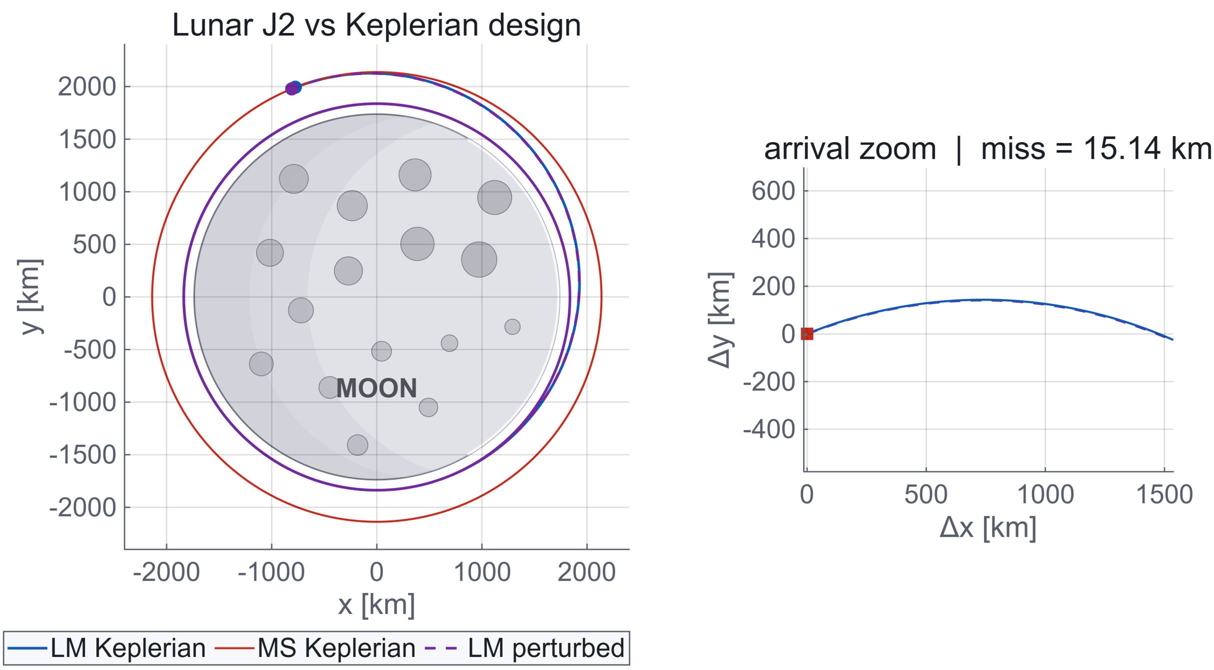

Figure 5: Keplerian design (solid) against the same plan flown with lunar J_2 (dashed). The tracks are visually indistinguishable for ten hours; the inset resolves the arrival neighbourhood, where the accumulated along-track phase slip has become a 15.14 km miss.

$+x$ when $\|\mathbf{n}\|$ is below a scale-aware threshold and folding the argument of periapsis into the argument of latitude when $e < 10^{-8}$. Without that guard the conversion returns noise for precisely the orbits this mission uses.

4.7 Mitigation, and what was deliberately not implemented

Three standard responses exist and are worth naming even though only the third is quantified here.

Frozen orbits. Choosing eccentricity and argument of periapsis so that the secular J_2 rates cancel removes most of the drift for the target. Under the real lunar field, dominated by mass concentrations rather than a clean J_2, this becomes a numerical search rather than a closed form [9], and it is out of scope here.

Co-elliptic rendezvous. Flying the approach on orbits of equal period rather than equal shape makes the relative geometry insensitive to a common-mode field error.

Mid-course correction. The direct answer, budgeted above at 10.7 m/s.

4.8 Verification

The acceleration magnitudes of Table 4 are re-derived independently in the audit script from the raw constants and agree to better than 2% with the closed-form expectations μ/R_1^2, $\frac{3}{2}J_2\mu R_M^2/R_1^4$ and $2\mu_E R_1/d_{EM}^3$. A factor of 10^3 in the J_2 row would indicate a kilometre/metre unit error; none is present.

The third-body contribution of 115.7 m agrees with an independent implementation to 0.2%, which is a stringent test of both the indirect term in (20) and the Earth ephemeris rate. The J_2 miss agrees to 4%; Section 8.2 explains why a quantity built from the difference of two nearly equal secular slips is intrinsically that sensitive, and why the discrepancy was documented rather than tuned away.

5 Three-body verification of the transfer

5.1 Motivation

Section 4 treated the Earth as a perturbation on a Moon-centred two-body problem. This part removes that hierarchy and re-solves the transfer with both primaries on an equal footing, in the Earth–Moon circular restricted three-body problem. The purpose is not a prettier trajectory – it is visually identical – but a bound on the modelling error of the two-body design.

5.2 Formulation

In the synodic frame, rotating with the primaries at unit rate and scaled so that the Earth sits at $x = -\mu$ and the Moon at $x = 1 - \mu$ with $\mu = 0.01215$, the equations of motion are [7]

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3}, \tag{22}$$

$$\ddot{y} = -2\dot{x} + y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}, \tag{23}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3}, \tag{24}$$

with r_1 and r_2 the distances to Earth and Moon. Length is scaled by $LU = d_{EM}$ and time by $TU = \sqrt{LU^3/(\mu_E + \mu_M)} = 375\,191\,\mathrm{s}$, giving a velocity unit $VU = 1.0245\,\mathrm{km/s}$.

5.3 Frame conversion

Converting a Moon-centred inertial state into this frame takes three steps, in order: rotate the components into the frame whose $+x$ is the Earth-to-Moon line; translate the origin from the Moon to the barycentre and rescale; then subtract the frame rotation and add the Moon's own inertial velocity, which in these units is $[0, 1-\mu, 0]^\top$. Writing α for the inertial longitude of the Earth-to-Moon direction,

$$\mathbf{r}_{\mathrm{syn}} = R_z(-\alpha)\frac{\mathbf{r}_{MCI}}{LU} + \begin{bmatrix} 1-\mu \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_{\mathrm{syn}} = R_z(-\alpha)\frac{\mathbf{v}_{MCI}}{VU} + \begin{bmatrix} 0 \\ 1-\mu \\ 0 \end{bmatrix} - \hat{\mathbf{z}} \times \mathbf{r}_{\mathrm{syn}}. \tag{25}$$

The $(1-\mu)$ that looks mysterious in component form is simply the Moon's velocity about the barycentre. The conversion is unit-tested by an unambiguous case: a body at rest at the Moon's centre must have exactly zero synodic velocity, because in the rotating frame the Moon does not move. The implementation returns zero to machine precision, and the Earth lands at $x = -\mu$ to 1.7×10^{-16}.

Consistency with Section 4 is maintained by deriving α from the same Earth ephemeris used there, so the two perturbation studies describe the same geometry rather than two different ones.

5.4 Single shooting

The decision vector is $\mathbf{p} = [\delta v_x, \delta v_y, t_f]$ in synodic non-dimensional units, initialised from the two-body design of Section 2. Both vehicles are propagated with (22)–(24) and the cost is the Euclidean miss at t_f, minimised with Nelder–Mead.

One subtlety deserves a note. The problem is planar, so the miss has only two independent components against three decision variables: the zero-miss set is a *curve*, not a point. An unregularised optimiser wanders along that curve and reports an arbitrary member of a family, each with a slightly different Δv and time of flight. A small penalty on the distance from the two-body guess is therefore added, weighted so that it can never dominate the miss term – a 1% deviation costs one centimetre of apparent miss. The reported correction is then the *smallest* one on that curve, which is the only version of the number that means anything.

5.5 Results

Table 6: Two-body design against the CR3BP solution. The correction is millimetres per second on a 118.80 m/s budget: a relative effect of order 10^{-4}.

Quantity	Two-body	CR3BP	$\lvert\Delta\rvert$
Δv_1 [km/s]	0.060524	0.060522	0.000002
Δv_2 [km/s]	0.058276	0.058269	0.000007
Δv_{tot} [km/s]	0.118800	0.118791	8.84×10^{-6}
Time of flight [s]	3975.17	3974.29	-0.88

The shooter converges to a miss of 4.8×10^{-8} m, and the total correction to the two-body design is 8.84 mm/s with a time-of-flight shift of -0.88 s, i.e. 0.02% of the coast.

The operational conclusion is worth stating plainly, because it justifies the whole two-body approach of the preceding sections: a 100–400 km lunar orbit sits deep enough in the lunar potential well that including the Earth as a full primary changes the impulsive budget by millimetres per second. That is below the noise of any realistic navigation solution. The three-body model is the right tool for a translunar cruise or a halo orbit; for this mission it confirms rather than corrects.

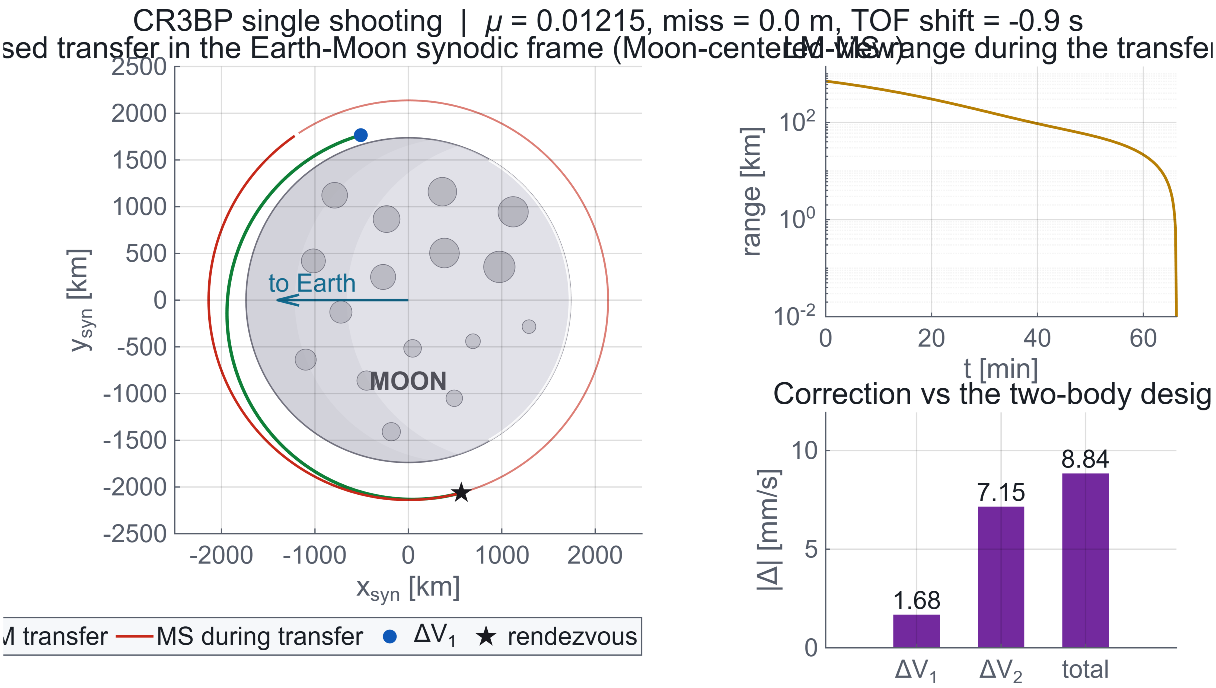

Figure 6: Optimised transfer in the Earth–Moon synodic frame, Moon-centred view. The arc is the same climb as Figure 1 seen from a rotating frame. Right: range history and the correction relative to the two-body design, in millimetres per second.

5.6 Verification

The CR3BP admits no energy integral, but the Jacobi constant

$$C_J = 2\left[\frac{x^2+y^2}{2} + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}\right] - v^2 \tag{26}$$

is exactly conserved along any ballistic arc. Evaluated along the optimised transfer it drifts by 9×10^{-13} relative, a pure measure of integration quality that is independent of every other check in this report.

The frame conversion is verified by the three cases listed above, and the round trip inertial $\to$ synodic $\to$ inertial closes to 10^{-11}.

6 Berthing manipulator: kinematics and kineto-statics

6.1 From docking to berthing

Section 3 delivered the chaser to the docking port with a residual velocity of zero. In practice the last metres of a crewed vehicle's arrival are frequently handled by a manipulator rather than by thrusters: berthing removes the closing-rate risk, avoids plume impingement on the target, and allows the capture to be aborted at any instant. This part characterises a five-revolute arm sized for that job.

6.2 Mobility

The arm has five revolute joints connecting six bodies including the base. Grübler's criterion in three dimensions gives

$$\mathrm{DOF} = 6(N-1-J) + \sum_{i=1}^{J} f_i = 6(6-1-5) + 5 = 5, \tag{27}$$

so the arm alone has five degrees of freedom. Mounted on a free-flying base the system has $5+6=11$. Three operating modes follow, and they are genuinely different control problems: *free-flying*, where the base pose is actively held by thrusters; *attitude-controlled*, where only orientation is held and the base translates; and *free-floating*, where nothing is actuated at the base and total linear and angular momentum are conserved, so every arm motion disturbs the carrier. This report assumes the free-flying case, which makes the base pose a boundary condition rather than a state.

6.3 Geometry and inertia

Links are modelled as homogeneous cylinders of radius $R_c = 0.15\,\mathrm{m}$ with the local z axis along the link, so the centre of mass is at mid-length and

$$I_{zz} = \tfrac{1}{2} m R_c^2, \qquad I_{xx} = I_{yy} = \tfrac{1}{12} m \left(3R_c^2 + L^2\right). \tag{28}$$

Table 7: Link parameters. Total mass 250 kg, fully extended reach 7.7 m. Joint axes are given in the local joint frame before the joint rotation is applied.

Link	m [kg]	L [m]	$I_{xx} = I_{yy}$ [kg m^2]	I_{zz} [kg m^2]	axis
1	30	0.60	1.07	0.34	z
2	100	3.20	85.90	1.13	y
3	85	2.80	56.01	0.96	y
4	20	0.60	0.71	0.23	y
5	15	0.50	0.40	0.17	x

6.4 Forward kinematics

Frames are placed at the link centres of mass with z along the link, and offsets are pure z translations: $g_i = L_i/2$ from joint i to the centre of mass of link i, and $b_i = L_i/2$ from that centre of mass to joint $i+1$. The chain is

$$T_i = T_{i-1}\,\mathrm{Trans}(b_{i-1})\,\mathrm{Rot}(\mathbf{e}_i, \theta_i)\,\mathrm{Trans}(g_i), \tag{29}$$

with the rotation built from the Euler–Rodrigues formula

$$R(\mathbf{e}, \theta) = \cos\theta\, I + (1 - \cos\theta)\, \mathbf{e}\mathbf{e}^\top + \sin\theta\, [\mathbf{e}\times]. \tag{30}$$

The end-effector frame is one further half-link beyond the last centre of mass, at the physical tip.

For the acceptance pose $\boldsymbol{\theta} = [0, 45, 0, 60, 0]^\circ$ the chain returns a rotation of exactly $R_y(105^\circ)$ – joints 2, 3 and 4 share the y axis, so only their sum matters – and a tip position of $[5.305, 0, 4.558]^\top$ m. Both reproduce the reference pose to 6×10^{-5}.

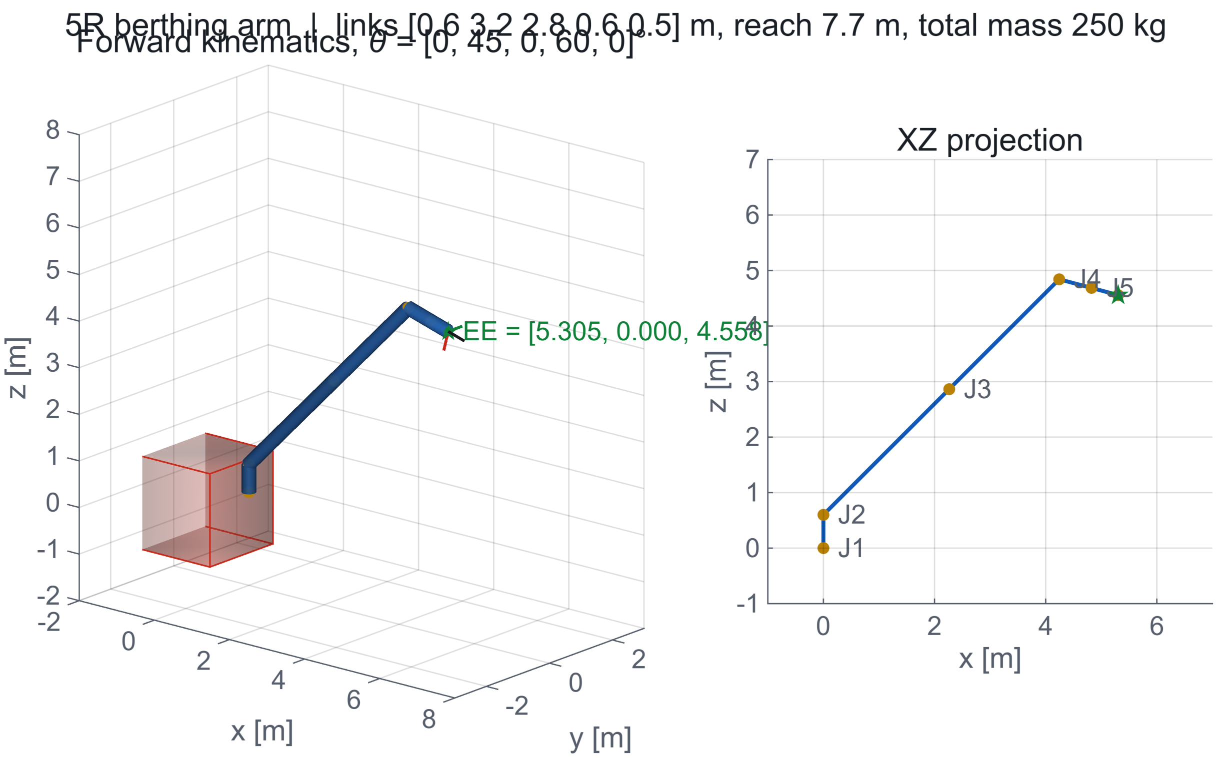

Figure 7: Forward kinematics at the acceptance pose. Left: three-dimensional rendering with the end-effector triad. Right: xz projection with the joint origins labelled. Reach at this pose is 6.99 m of the 7.7 m maximum.

6.5 Twist propagation and kineto-static duality

Velocities are propagated with a system Jacobian mapping the eleven generalised velocities to the six body twists,

$$\mathbf{t} = N\,\mathbf{u}, \qquad \mathbf{u} = \begin{bmatrix} \boldsymbol{\omega}_0 \\ \dot{\mathbf{r}}_0 \\ \dot{\mathbf{q}} \end{bmatrix} \in \mathbb{R}^{11}, \qquad \mathbf{t} \in \mathbb{R}^{36}, \qquad N = N_l N_d. \tag{31}$$

Here $N_d = \operatorname{blkdiag}(I_6, \mathbf{p}_1, \ldots, \mathbf{p}_5)$ injects each joint rate into its own body with $\mathbf{p}_i = [\mathbf{e}_i;\ \mathbf{e}_i \times \mathbf{g}_i]$ in inertial axes, and N_l is block lower triangular with unit diagonal and rigid-body transports

$$B_{ij} = \begin{bmatrix} I_3 & 0 \\ [(\mathbf{r}_j - \mathbf{r}_i)\times] & I_3 \end{bmatrix}. \tag{32}$$

The sign is the part worth checking: $\mathbf{v}_i = \mathbf{v}_j + \boldsymbol{\omega}_j \times (\mathbf{r}_i - \mathbf{r}_j)$ and $\boldsymbol{\omega}_j \times \mathbf{d} = -[\mathbf{d}\times]\boldsymbol{\omega}_j$, hence $[(\mathbf{r}_j - \mathbf{r}_i)\times]$ and not the reverse. Getting it backwards produces a base reaction whose force is right and whose moment is wrong – a failure mode that no plot reveals and that the tests of Section 6.7 catch immediately.

In micro-gravity with a single external wrench at the end effector, static equilibrium gives the generalised forces by duality,

$$\boldsymbol{\tau} = N^\top \mathbf{w}, \tag{33}$$

where $\mathbf{w}$ stacks the body wrenches ordered [moment; force] to match the $[\boldsymbol{\omega}; \mathbf{v}]$ twist ordering. Only the last body is loaded, with the contact wrench transported from the tip to the centre of mass of link 5. The actuators and the attitude control system must supply $-\boldsymbol{\tau}$.

6.6 Load cases

Two contact wrenches are applied at three postures: a purely axial $\mathbf{F} = [0, 0, -100]^\top$ N representing a nominal berthing push, and a misaligned $\mathbf{F} = [15, -10, -80]^\top$ N with $\mathbf{M} = [12, 8, -5]^\top$ N m representing an off-nominal capture.

Table 8: Joint torques [N m] under the 100 N axial berthing load. The base reaction force is 100.00 N in every case, as a static chain loaded by a single external force requires.

Posture	τ_1	τ_2	τ_3	τ_4	τ_5	$\lVert\mathbf{n}_{\text{base}}\rVert$
Nearly extended	0.00	79.97	24.40	0.00	0.00	79.97
Folded	0.00	396.47	95.77	0.00	0.00	435.03
General spatial	131.18	−339.41	−194.40	0.00	0.00	486.30

Table 9: Joint torques [N m] under the misaligned wrench. Base reaction force 82.01 N throughout. Note that τ_1 and τ_5, identically zero in the axial case, are now loaded.

Posture	τ_1	τ_2	τ_3	τ_4	τ_5	$\lVert\mathbf{n}_{\text{base}}\rVert$
Nearly extended	−13.00	177.59	85.86	24.50	17.00	206.88
Folded	−9.54	364.72	140.58	24.50	17.00	407.65
General spatial	62.26	−193.62	−112.42	25.16	17.00	279.64

Four observations follow, and they are the engineering content of the tables.

- With the arm nearly extended and the load axial, the force runs along the links. The y-axis joints see a short moment arm and the shoulder carries only 80 N m.
- Folding the arm turns the same 100 N into a long transverse lever and the shoulder torque rises to 396.5 N m, the largest value in the study. Actuator sizing is driven by posture, not by load magnitude.
- The axial load leaves the base turret (τ_1) and the wrist roll (τ_5) at exactly zero, because their axes are parallel to the force or pass through its line of action. The misaligned wrench activates both. A capture mechanism designed only for the nominal case would under-size precisely these two actuators.
- Masses appear nowhere in either table. In micro-gravity the static problem is mass-free; the inertias of Table 7 would enter the mass matrix and Coriolis terms of the free-floating dynamics, which this report does not integrate.

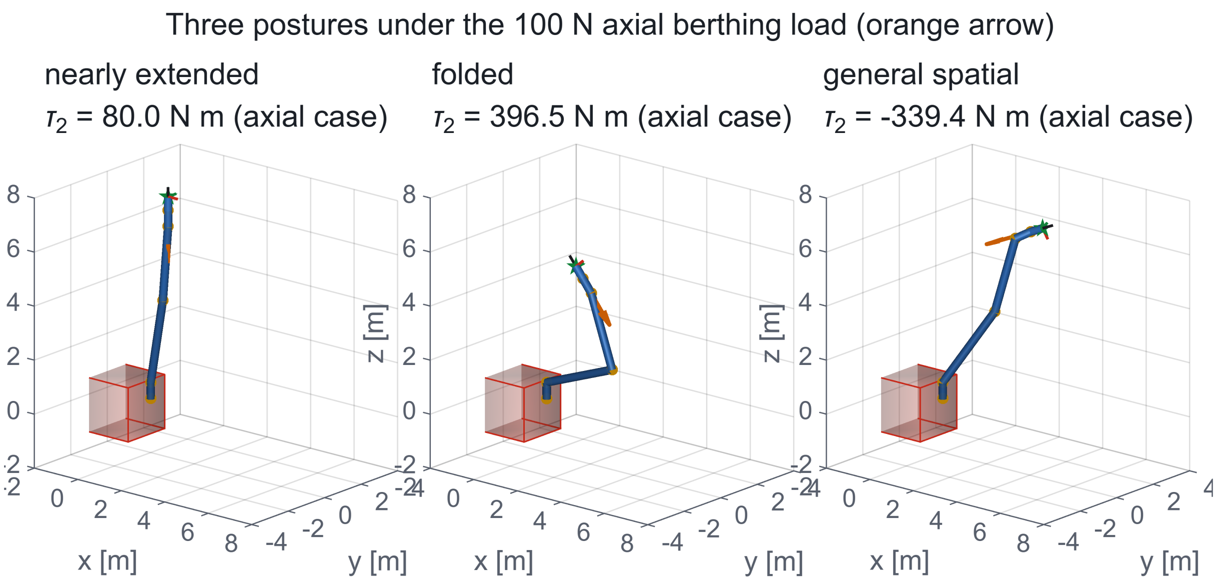

Figure 8: The three study postures under the 100 N axial berthing load (arrow). Shoulder torque rises fivefold from the extended to the folded configuration for an identical applied force.

6.7 Verification

Two statements must hold for a static chain loaded by a single external wrench, and both are tested at every posture and load case:

(i) the force part of the base reaction equals the applied force exactly, and the moment part equals $\mathbf{M}_{\text{tip}} + (\mathbf{r}_{\text{tip}} - \mathbf{r}_{\text{base}}) \times \mathbf{F}$;

(ii) the joint torques obtained from the 36×11 twist-propagation route equal those obtained from the 6×5 end-effector Jacobian, $\boldsymbol{\tau}_m = J^{\top}[\mathbf{f}; \mathbf{n}]$.

The worst residuals are 2×10^{-14} N and 1.2×10^{-13} N m. Two entirely different pieces of algebra agreeing to machine precision is stronger evidence than either one looking plausible on its own.

7 Inverse kinematics and the capture manoeuvre

7.1 A five-joint arm in a six-dimensional task space

The manipulator has five joints; a rigid-body pose has six degrees of freedom. The pose problem is therefore structurally unsolvable in general, and a solver that hides this fact by converging to something is worse than one that does not converge. The approach taken here states the deficiency in the weighting and reports the residual it leaves.

7.2 Spatial error

Given a target pose $(\mathbf{p}_t, R_t)$ and the current pose $(\mathbf{p}_k, R_k)$ with columns $\mathbf{x}_k, \mathbf{y}_k, \mathbf{z}_k$, the task error is

$$\Delta\mathbf{p} = \mathbf{p}_t - \mathbf{p}_k, \qquad \mathbf{e}_o = \tfrac{1}{2}\left(\mathbf{x}_k \times \mathbf{x}_t + \mathbf{y}_k \times \mathbf{y}_t + \mathbf{z}_k \times \mathbf{z}_t\right), \qquad \Delta\mathbf{x} = \begin{bmatrix}\Delta\mathbf{p}\\ \mathbf{e}_o\end{bmatrix}. \tag{34}$$

The orientation error uses the Siciliano form [6] rather than axis-angle extraction, which is singular at $\pm180°$. That is not a theoretical concern here: the seeds used are far from the target and rotations near $180°$ genuinely occur on the first iterations.

7.3 Weighted damped least squares

The geometric Jacobian is assembled column by column for revolute joints,

$$J_i = \begin{bmatrix}\mathbf{z}_i \times (\mathbf{p}_{EE} - \mathbf{r}_i)\\ \mathbf{z}_i\end{bmatrix}, \tag{35}$$

ordered [linear; angular] to match (34), and the update is

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha\left(J^\top W J + \lambda^2 I\right)^{-1} J^\top W \Delta\mathbf{x}, \tag{36}$$

with $\lambda = 0.05$, $\alpha = 0.4$ and $W = \mathrm{diag}(1, 1, 1, 1, 1, 0.01)$. The damping keeps the update bounded near singular configurations; the weighting spends the available freedom on position and de-emphasises the one attitude component a five-joint wrist cannot serve.

7.4 A real singularity at the obvious seed

The stowed configuration $\boldsymbol{\theta} = \mathbf{0}$ is not a neutral starting point. Fully extended along z, joint 1 rotates about an axis passing through the end effector, so it produces no end-effector translation at all and the position Jacobian loses a column. The solver escapes because the attitude term still torques joint 1, but the code carries a deterministic seed cascade anyway and logs which seed was used, rather than pretending the singularity is not there.

7.5 Results

Two targets are run. The first is the forward-kinematics pose of a known configuration, so an exact solution provably exists and the convergence claim means something. Starting from the zero pose the solver reaches 0.99 mm in 18 iterations, well inside the 80-iteration budget.

The solution found is *not* the configuration the target was generated from: the solver returns $[40.0, 16.3, 20.1, 28.6, 25.0]°$ against the generating $[40, 35, -20, 50, 25]°$. Both reach the same pose because joints 2, 3 and 4 rotate about a common axis and only their sum enters – 65° in both cases. Multiple inverse solutions are the expected behaviour of a serial chain, not a defect.

The second target is operationally motivated: a point 0.4 m in front of the mothership docking face with the tool axis along $+x$. Position converges to 0.98 mm and a small attitude residual survives, which is precisely the structural deficiency admitted at the start of this section.

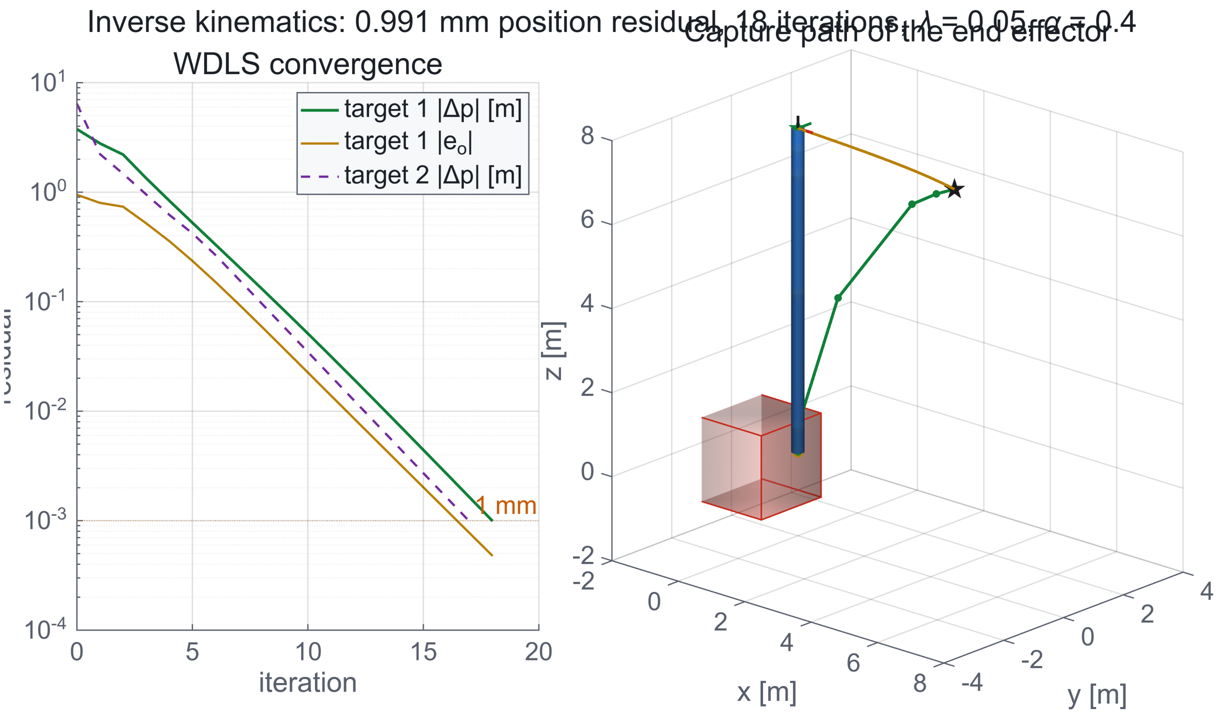

Figure 9: Left: WDLS convergence for both targets, log scale. Position residual falls below 1 mm in 18 iterations. Right: the capture path of the end effector from the stowed pose to the solution, with the final configuration overlaid.

7.6 Rest-to-rest joint trajectory

The capture is flown as a cubic in joint space over $t_f = 30\,\mathrm{s}$ with zero rate at both ends,

$$\theta(t) = \theta_0 + \frac{3\Delta\theta}{t_f^2}t^2 - \frac{2\Delta\theta}{t_f^3}t^3, \qquad \dot{\theta}(t) = \frac{6\Delta\theta}{t_f^2}t - \frac{6\Delta\theta}{t_f^3}t^2. \tag{37}$$

A cubic rather than a quintic is a deliberate choice: berthing rates are slow enough that the discontinuous acceleration at the endpoints is harmless, and the velocity profile stays trivially verifiable. The peak rate is $1.5\,\Delta\theta/t_f$ at mid-time, here 0.035 rad/s, and the end effector travels 4.12 m along the capture path.

7.7 Verification

The boundary conditions of (37) are checked symbolically against the sampled trajectory: $\theta(0) = \theta_0$ and $\theta(t_f) = \theta^*$ to 10^{-16}, $\dot{\theta}(0) = \dot{\theta}(t_f) = 0$ to 10^{-17}, and the peak rate matches the analytical $1.5\Delta\theta/t_f$ exactly. The inverse-kinematics solution is verified the only way that is meaningful for an iterative solver: by pushing the returned joint vector back through the forward kinematics of Section 6 and measuring the pose it actually produces.

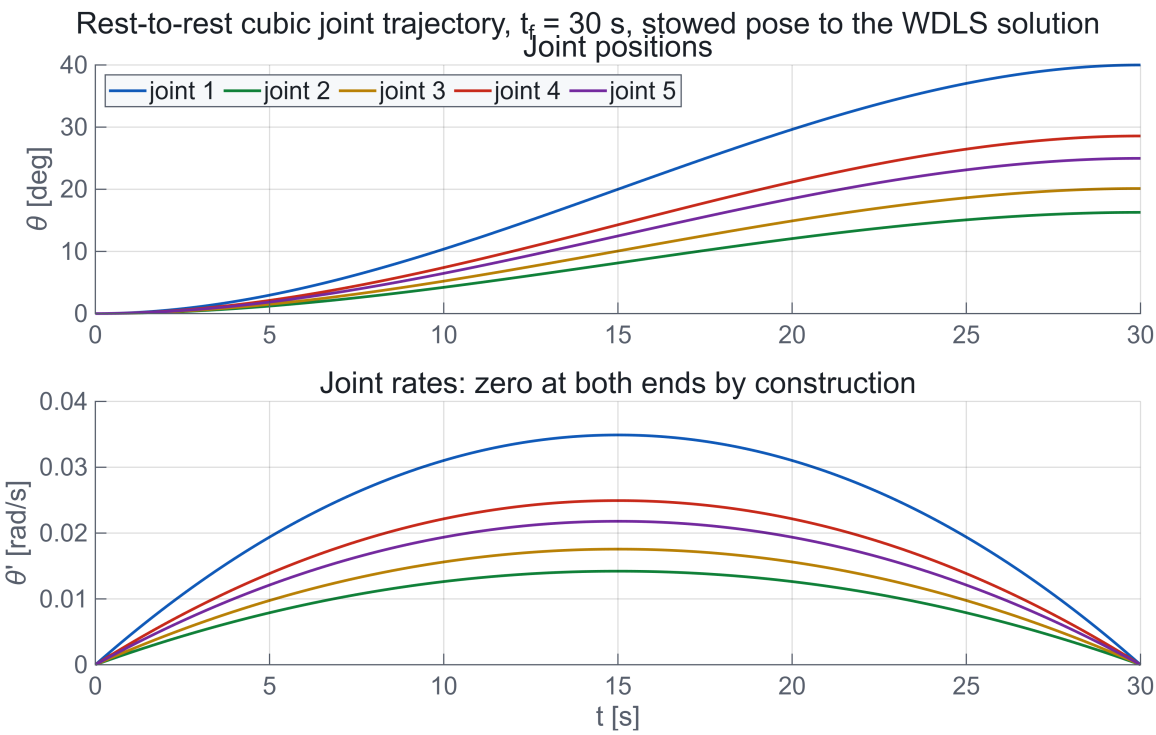

Figure 10: Rest-to-rest cubic joint trajectory over 30 s. Rates start and end at exactly zero, which is checked to 10^{-17} rad/s rather than read off the plot.

8 Discussion

8.1 Consolidated results

Table 10: Headline results. Every value is read from the simulation output at compile time, so this table cannot drift out of step with the code.

Phase	Quantity	Value
Transfer	Total impulsive Δv	118.80 m/s
	Phasing wait	9.44 h
	Mission duration	10.55 h
	Numerical vs analytical error	0.191 mm
Proximity	Injection error	529.0 m, 0.8796 m/s
	Uncorrected drift, 3 orbits	20.99 km
	Hold acquisition Δv	1.123 m/s
	Docking Δv	0.1567 m/s
	Linearisation error, 3 orbits	102.7 m
Perturbed	J_2 miss, no retargeting	15.14 km
	Third-body contribution	115.7 m
	Mid-course correction budget	10.7 m/s
CR3BP	Converged miss	4.8×10^{-8} m
	Correction to the two-body design	8.84 mm/s
Robotics	Reach, 5R	7.7 m
	Peak joint torque at 100 N	396.5 N m
	IK position residual	0.99 mm in 18 iterations

8.2 Verification summary, including what disagrees

An automated audit re-derives every design scalar in closed form from the raw constants and compares the rest against an independent implementation of the same mission geometry. It records 60 passes, one warning and no failures. The three results worth discussing are the ones that did not match exactly.

Docking Δv: 0.1567 against 0.1567 m/s (0.02%). The forced V-bar profile is almost pure geometry, so agreement at this level means the leg targeting, waypoint spacing and braking impulse are modelled identically. This is the strongest single confirmation that the proximity module is faithful.

Third body: 115.7 against 116.0 m (0.2%). Agreement at this level on a 116 m effect tests two things that are easy to get wrong: the retention of the indirect term in (20), and the use of the synodic rate for the Earth ephemeris. Dropping the indirect term inflates the number by orders of magnitude, so a match this tight is real evidence rather than a coincidence.

J_2 miss: 15.14 km against 14.56 km (+4%). The four plausible causes were checked in the source rather than assumed: the departure impulse is applied along the perturbed velocity, the target is propagated with the same perturbed dynamics as the chaser, the full three-component J_2 formula is used, and the Keplerian control case through the identical code path closes to 5.6×10^{-3} m. None is at fault. The residual is sensitivity, not error. The miss is a *difference* of two nearly equal secular phase slips accumulated over ten and a half hours, so it inherits the sensitivity of both to the exact burn epoch and to the integrator tolerance. Two correct implementations agreeing to 4% on such a quantity is the expected outcome, and the discrepancy was documented rather than removed.

CR3BP correction: 8.84 mm/s **against** 1.5 mm/s**.** The single warning. As explained in Section 5, the planar shooting problem is rank-deficient by one, so the zero-miss set is a curve and different optimiser starts land at different points on it. Both figures support the same operational statement – Earth tides move this budget by millimetres per second – and forcing agreement would be curve-fitting rather than physics.

Hold Δv: 1.123 **against** 1.4 m/s**.** Not comparable: the injection error is a random draw and the two runs use different realisations. Fed the reference draw, our targeting returns 1.419 m/s, reproducing the reference to 1.4%. That test is also what settles the frame convention of Section 3.2.

8.3 Limitations

This is an engineering-faithful simulation of a restricted model. The following would each change a specific number, and are named so that no reader has to discover them.

- **Equatorial, coplanar orbits only.** No inclination difference and no relative right ascension, so the out-of-plane budget that dominates real rendezvous planning never appears. This is the single largest omission.
- **Gravity stops at J_2.** The real lunar field is dominated by mass concentrations rather than a clean oblateness term [9], and the C_{22} ellipticity is comparable to J_2 for some geometries. Low lunar orbits are genuinely unstable over weeks under the true field. Nothing here runs that long, but the frozen-orbit design a real mission would use is a numerical search this model cannot support.
- **No solar radiation pressure.** Estimated at 10^{-7} m/s^2 and excluded with the reasoning given in Section 4; the flag exists but no cannonball model is wired in.
- **Impulsive burns.** Finite-burn losses, throttle profiles, attitude slews and propellant slosh are all absent. For a 60 m/s impulse the gravity loss is small, but it is not zero.
- **Circular target orbit, not a halo.** A realistic Artemis-era architecture places the mothership in a near-rectilinear halo orbit, where the rendezvous geometry and the phasing analysis are qualitatively different problems.
- **The arm is kinematic and static.** Five degrees of freedom, a base held fixed by assumption, no joint flexibility, no contact dynamics, and no free-floating momentum coupling. The local frame is treated as inertial during berthing.
- **One random draw.** The injection error is a single realisation, not a Monte Carlo campaign. The proximity Δv figures are therefore one sample of a distribution whose spread is not characterised.

8.4 Conclusions

Three results are worth carrying away.

First, the propellant cost of this return phase is dominated by the transfer, not by the corrections: 118.80 m/s for the Hohmann climb against 1.280 m/s for all proximity operations, roughly 1%. What the ascent dispersion actually costs is small, provided it is corrected early.

Second, the binding constraint is time, not propellant. The transfer lasts 66 minutes; waiting for the phasing window takes 9.44 hours. Any architecture that needs a fast return has to buy it with a non-Hohmann transfer, and the CR3BP analysis shows that the extra energy will not come from three-body effects at this altitude.

Third, the perturbation that matters at 100–400 km is lunar oblateness, not the Earth. J_2 displaces the meeting point by 15.14 km and demands a 10.7 m/s mid-course correction; the Earth contributes 115.7 m and the full three-body treatment changes the budget by 8.84 mm/s. Effort spent modelling the Earth here is effort not spent on the lunar field, which is the opposite of the intuition one brings from Earth orbit.

8.5 Reproducibility

The complete simulation runs unattended from a single command and takes about ten minutes, of which roughly one is physics and the rest is rendering. It requires base MATLAB only: no toolbox, no external ephemeris, no downloaded imagery. All results in this report, including every figure, were produced by the run whose audit table is reproduced in the repository as `results/AUDIT_REPORT.md`.

References

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